EMPLOYEE PERFORMANCE PREDICTION

USING MACHINE LEARNING

SYBCA-AI (Semester III)

303 Machine Learning - Mini Project

Supervised Learning – Multiclass Classification

Submitted by: _____

Roll Number: _____

Submitted to: _____

Academic Year: 2026–2027

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his/her Practical work in Paper No. **303-05 Fundamentals of Machine Learning (PR)** *for the*

Semester **III** *during the academic year* **2026-27**

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Examiner : 12/10

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1. Problem Statement

Employee performance can be represented using workplace and employee-related attributes such as attendance, experience, training, completed projects, salary and job satisfaction. This project builds a supervised multiclass classification model that predicts an employee's performance category.

2. Objectives

• Predict employee performance using supervised machine learning.

• Analyze relationships among attendance, experience, training, projects, salary and job satisfaction.

• Convert categorical data into machine-readable form and scale numerical features.

• Train and evaluate a multiclass Logistic Regression model.

• Present findings using charts and evaluation metrics.

3. Dataset Description

Column	Type	Description
Employee_ID	Categorical	Unique employee identifier
Age	Numeric	Employee age
Gender	Categorical	Gender category
Department	Categorical	Department of employee
Education	Categorical	Education level
Experience	Numeric	Years of experience
Attendance	Numeric	Attendance percentage
Working_Hours	Numeric	Average working hours
Training_Hours	Numeric	Training hours
Projects_Completed	Numeric	Completed projects
Overtime	Categorical	Overtime status
Salary	Numeric	Salary amount
Job_Satisfaction	Numeric	Satisfaction score 1–5
Performance_Rating	Target	Poor / Average / Good / Excellent

4. Data Preprocessing

The supplied dataset contains 1000 records and 14 columns. No missing values were found and no duplicate rows were found. Employee_ID was excluded from model training because it is an identifier rather than a predictive feature.

Preprocessing Steps

• Remove Employee_ID from model features.

• Separate input features (X) and target variable (y).

• Standardize numerical features using StandardScaler.

• Convert Gender, Department, Education and Overtime using OneHotEncoder.

• Split the dataset into 80% training and 20% testing data.

• Use stratification so all four target classes remain represented.

Workflow: CSV → Feature/Target Split → Train/Test Split → Scaling + One-Hot Encoding → Logistic Regression → Prediction → Evaluation

A preprocessing pipeline applies the same transformations consistently before classification.

5. Exploratory Data Analysis

The dataset contains 361 Good, 294 Excellent, 272 Average and 73 Poor records. The classes are not perfectly balanced, with Poor being the smallest class.

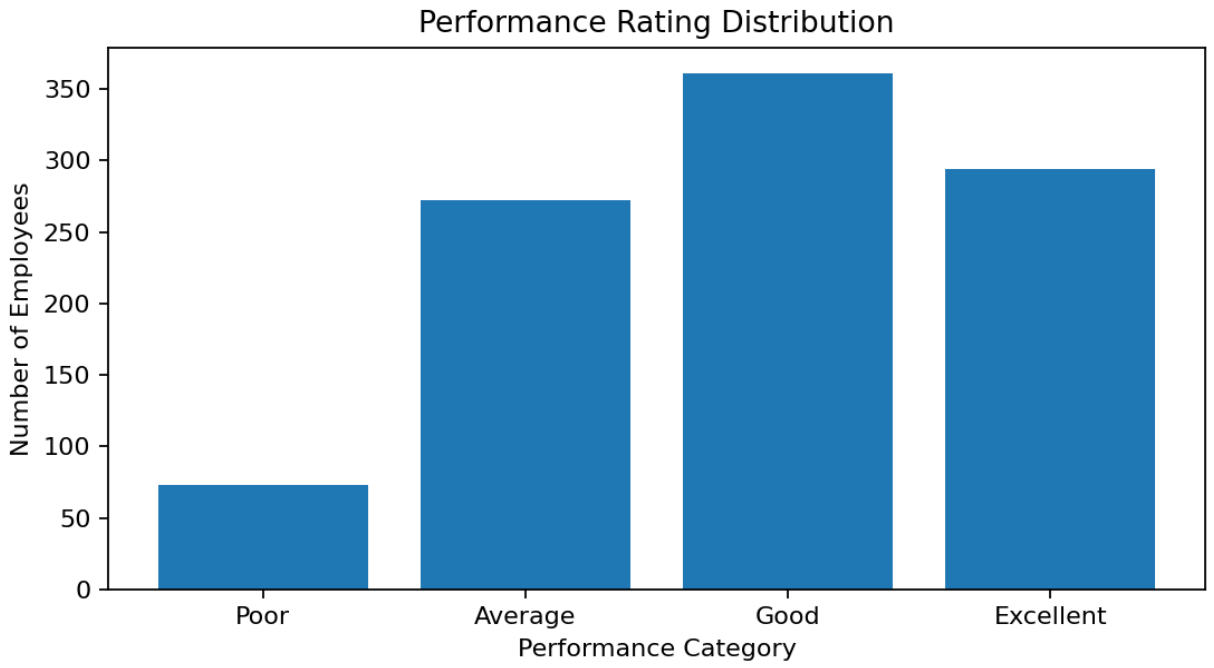


Figure 1: Performance rating distribution

The distribution chart shows the relative number of records in each target category before model training.

5.1 EDA Findings

The supplied report describes an attendance-versus-encoded-performance chart for exploratory understanding. The report does not provide the employee-level attendance values required to reconstruct that exact chart, so no invented attendance observations are included here.

Target distribution used in the analysis

Performance Category	Records	Share
Poor	73	7.3%
Average	272	27.2%
Good	361	36.1%
Excellent	294	29.4%
Total	1000	100%

The source report specifically notes that the attendance chart is for exploratory understanding and does not by itself prove causation.

6. Model Building and Feature Selection

Logistic Regression was selected for multiclass classification. The model learns patterns from the training data and predicts one of four performance categories for unseen employees.

Model Workflow

• Load CSV dataset.

• Separate input features and target.

• Remove Employee_ID.

• Split data into training and testing sets (80:20).

• Scale numerical features.

• One-hot encode categorical features.

• Train Logistic Regression.

• Predict test-set performance.

• Evaluate using accuracy, precision, recall, F1-score and confusion matrix.

Model Pipeline

Numerical features → StandardScaler; Categorical features → OneHotEncoder; Transformed features → Logistic Regression → Performance_Rating prediction.

7. Model Evaluation and Results

Test-set accuracy: **94.50%** (200 test records).

Class	Precision	Recall	F1-score	Support
Poor	0.92	0.80	0.86	15
Average	0.90	0.96	0.93	54
Good	0.96	0.94	0.95	72
Excellent	0.98	0.97	0.97	59
Macro Avg	0.94	0.92	0.93	200
Weighted Avg	0.95	0.94	0.94	200

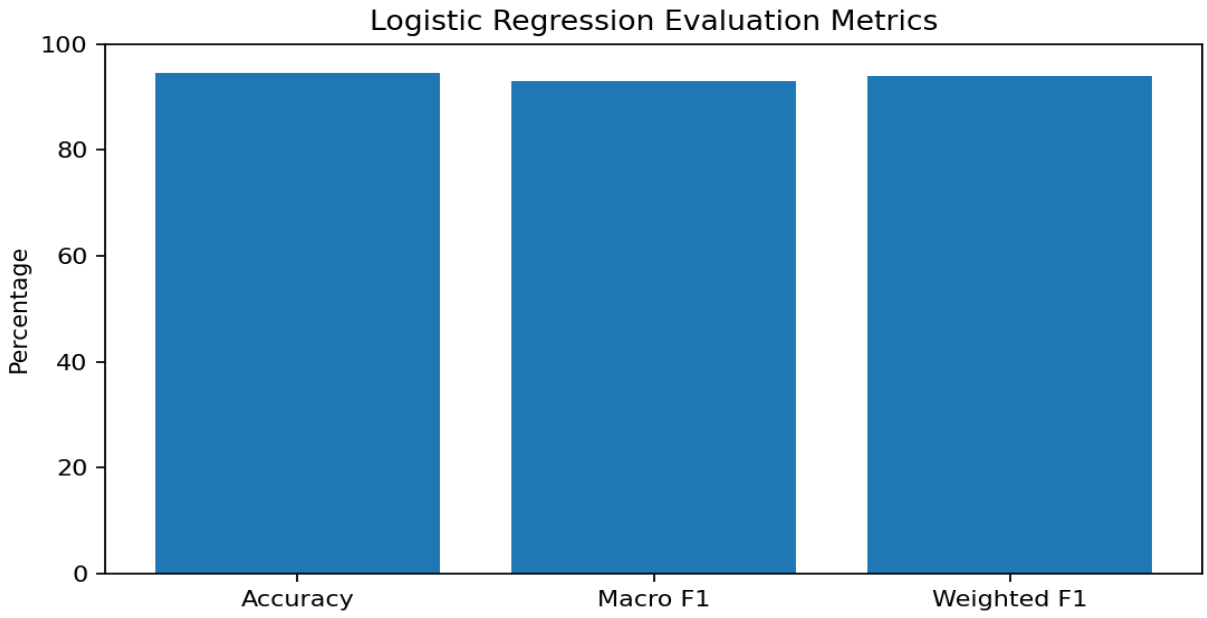


Figure 2: Reported Logistic Regression evaluation metrics

7.1 Results Interpretation

The Logistic Regression model achieved 94.50% accuracy on the held-out test set. The class-wise results provide precision, recall and F1-score for all four performance categories.

Key Observations

• Excellent has the highest reported F1-score at 0.97.

• Good has a reported F1-score of 0.95.

• Average has a reported F1-score of 0.93.

• Poor has a reported F1-score of 0.86 and is also the smallest class.

• The evaluation is based on the 200-record held-out test set.

These values describe model performance on this dataset and test split. They do not establish that any single employee attribute causes performance outcomes.

8. Conclusion

The Employee Performance Prediction model demonstrates a complete supervised machine learning workflow on the provided 1000-record dataset. Logistic Regression achieved 94.50% accuracy on the held-out test set and classified employees into Poor, Average, Good and Excellent performance categories based on the available attributes.

For real organizational deployment, the model should be validated on newer data and checked for fairness, data quality and business requirements.

9. Future Scope

• Compare Logistic Regression with Decision Tree, Random Forest and other suitable classifiers.

• Tune hyperparameters and use cross-validation.

• Add a user interface for entering employee details and obtaining a prediction.

• Monitor model performance on new data.

• Generate automated EDA/Sweetviz reports for future datasets.

10. Software and Libraries

Python, Pandas, NumPy, Matplotlib, Scikit-learn and ReportLab. The project can be implemented in VS Code, Jupyter Notebook or a similar Python environment.

11. References

1. Employee performance dataset: employee_performance_1000_records.csv.

2. Scikit-learn documentation – preprocessing, Logistic Regression and model evaluation.

3. Pandas documentation – data loading and analysis.

4. NumPy documentation – numerical operations.

5. Matplotlib documentation – data visualization.

Project Summary

Item	Details
Project	Employee Performance Prediction
Learning Type	Supervised Learning
Problem Type	Multiclass Classification
Dataset Size	1000 records
Input Features	13
Target Classes	4
Model	Logistic Regression
Test Records	200
Accuracy	94.50%

— End of Project Report —